Section I, Part A

1 If $g(x) = x^2\sqrt{x+1}$, then $g'(3) =$

(单选题)

A . $\frac{3}{2}$

B . $\frac{39}{4}$

C . $\frac{57}{4}$

D . 36

正确答案：C

解析

Answer C

Correct. The derivative of g is found using the product rule. $g'(x) = 2x \cdot \sqrt{x+1} + x^2 \cdot \frac{1}{2\sqrt{x+1}}$

$g'(3) = 6(2) + \frac{9}{2(2)} = 12 + \frac{9}{4} = \frac{57}{4}$

2 $\int_0^4 100e^{\frac{x}{4}}\,dx =$

（单选题）

A $.25(e-1)$

B $.100(e-1)$

C $.400(e-1)$

D $\cdot 400(e^4-1)$

正确答案：C

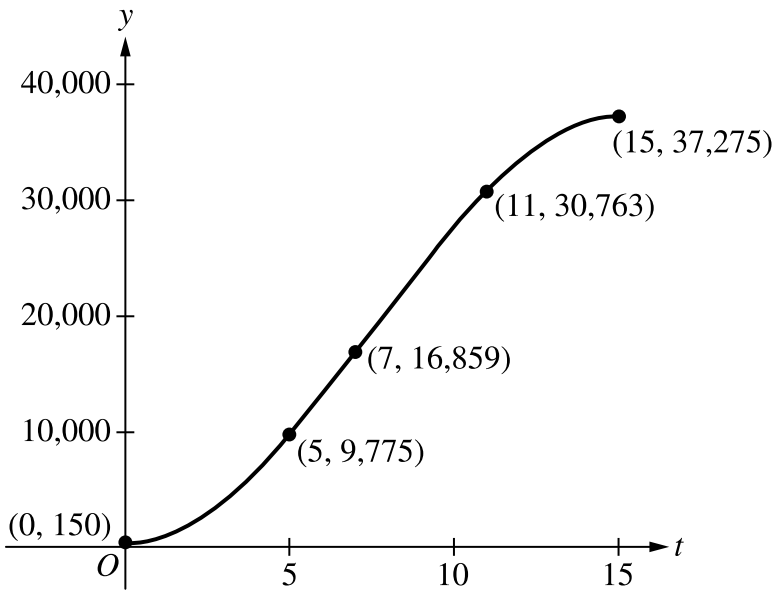
解析

Answer C

Correct. Starting with the substitution $u = \frac{x}{4}$, $u = \frac{x}{4} \Rightarrow \frac{du}{dx} = \frac{1}{4} \Rightarrow dx = 4\,du$. When $x = 0$, $u = 0$. When $x = 4$, $u = 1$. Substituting for $\frac{x}{4}$, for dx, and for the limits, $\int_0^4 100e^{\frac{x}{4}}\,dx = \int_0^1 100e^u(4\,du) = 400\int_0^1 e^u\,du = 400e^u\Big|_{u=0}^{u=1} = 400e - 400 = 400(e-1)$. Alternatively, the antiderivative of $e^{\frac{x}{4}}$ is $e^{\frac{x}{4}}$ with an adjustment of dividing by $\frac{1}{4}$ to account for the chain rule. Therefore,

$$\int_0^4 100e^{\frac{x}{4}}\,dx = 100\cdot\frac{e^{\frac{x}{4}}}{\frac{1}{4}}\Bigg|_{x=0}^{x=4} = 400e^{\frac{x}{4}}\Big|_{x=0}^{x=4} = 400e - 400 = 400(e-1).$$

3



An airplane's height above sea level is modeled by the differentiable function $y = h(t)$, where t is the number of minutes after takeoff and $h(t)$ is measured in feet. The graph of h is shown in the figure for $0 \le t \le 15$. Of the following, which is the best estimate of the rate at which the plane is rising above sea level, in feet per minute, at time $t = 9$?

(单选题)

A . $\frac{37,275-150}{15-0}$

B . $\frac{37,275-9,775}{15-5}$

C . $\frac{37,275-16,859}{15-7}$

D . $\frac{30,763-16,859}{11-7}$

正确答案：D

解析

Answer D

Correct. The symmetric difference quotient on the interval [7, 11] is used to approximate the instantaneous rate of change at $t = 9$. $\frac{h(11) - h(7)}{11 - 7} = \frac{30{,}763 - 16{,}859}{11 - 7}$ The average rate of change on the interval [7, 11] is the slope of the secant line between the points at $t = 7$ and $t = 11$. The shape of the graph suggests that this secant line is almost parallel to the line tangent to the graph at time $t = 9$. Therefore, the slope of this secant line is a good approximation to the slope of the tangent line.

4 Let f be a vector-valued function with $f(0) = \langle -2, 3 \rangle$. If the instantaneous rate of change of f with respect to t is given by $\langle 3t^2 + 4t + 1, 2t^3 + t - 2 \rangle$, what is $f(1)$?

(单选题)

A .$\langle 2, 2 \rangle$

B .$\langle 4, -1 \rangle$

C .$\langle 8, 10 \rangle$

D .$\langle 10, 7 \rangle$

正确答案：A

解析

Answer A

Correct. Let $f(t) = \langle x(t), y(t) \rangle$. Then $x'(t) = 3t^2 + 4t + 1$ and $y'(t) = 2t^3 + t - 2$. The Fundamental Theorem of Calculus is used with each component separately.

$$x(1) = x(0) + \int_0^1 x'(t)\,dx = x(0) + \int_0^1 (3t^2 + 4t + 1)\,dt = -2 + (t^3 + 2t^2 + t)\Big|_0^1 = -2 + (4 - 0) = 2$$

$$y(1) = y(0) + \int_0^1 y'(t)\,dt = y(0) + \int_0^1 (2t^3 + t - 2)\,dt = 3 + \left(\frac{1}{2}t^4 + \frac{1}{2}t^2 - 2t\right)\Big|_0^1 = 3 + (-1 - 0) = 2$$

5 $\lim\limits_{x \to 4} \dfrac{2 - \sqrt{x}}{x - 4}$ is

(单选题)

A . $-\dfrac{1}{4}$

B .0

C . $\dfrac{1}{4}$

D .nonexistent

正确答案：A

解析

Answer A

Correct. The given expression is multiplied by $\frac{2+\sqrt{x}}{2+\sqrt{x}}$ to rearrange the given expression into an equivalent form: $\frac{2-\sqrt{x}}{x-4} = \frac{2-\sqrt{x}}{x-4}\frac{2+\sqrt{x}}{2+\sqrt{x}} = \frac{4-x}{(x-4)(2+\sqrt{x})} = \frac{-1}{2+\sqrt{x}}$. It follows that $\lim_{x\to 4}\frac{-1}{2+\sqrt{x}} = -\frac{1}{4}$. Alternatively, after the minus sign has been factored out in the numerator, the limit can be written in the form of the difference quotient $\lim_{x\to 4}\frac{f(x)-f(4)}{x-4}$, where $f(x) = \sqrt{x}$. This limit is one way to express the derivative of f at $x = 4$. Since $f'(x) = \frac{1}{2\sqrt{x}}$, $\lim_{x\to 4}\frac{f(4)-f(x)}{x-4} = -\left(\lim_{x\to 4}\frac{f(x)-f(4)}{x-4}\right) = -f'(4) = -\frac{1}{2\sqrt{4}} = -\frac{1}{4}$.

6 Let $y = f(x)$ be the solution to the differential equation $\frac{dy}{dx} = x + y^2$ with initial condition $f(1) = 2$. What is the approximation for $f(1.4)$ if Euler's method is used, starting at $x = 1$ with two steps of equal size?

(单选题)

A .2.4

B .3.04

C .4

D .5.04

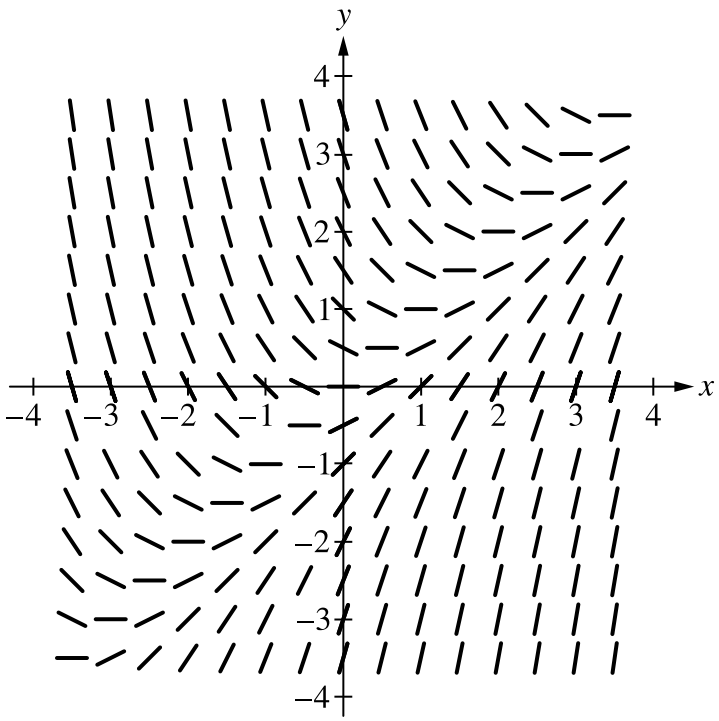
正确答案：D

解析

Answer D

Correct. Euler's method for the differential equation $\frac{dy}{dx} = f(x, y) = x + y^2$ can be written as $y_{k+1} = y_k + f(x_k, y_k)\,\Delta x$, where Δx is the step size. Then y_k is an approximation for $f(x_k)$. Here the step size is $\Delta x = \frac{1.4-1}{2} = 0.2$ since there are two steps of equal length. $x_0 = 1; y_0 = 2\, x_1 = 1.2; y_1 = y_0 + f(x_0, y_0)\,\Delta x = 2 + (1 + 4)(0.2) = 2 + 1 = 3$ $x_2 = 1.4; y_2 = y_1 + f(x_1, y_1)\,\Delta x = 3 + (1.2 + 9)(0.2) = 3 + 10.2(0.2) = 5.04$

7



A portion of the slope field for the differential equation $\frac{dy}{dx} = x - y$ is shown in the figure. If the linear function $y = f(x)$ is the solution to the differential equation such that $f(-1) = -2$, then $f(3) =$

(单选题)

A .0

B .1

C .2

D .3

正确答案：C

解析

Answer C

Correct. To estimate the value of $f(3)$ graphically, sketch the graph of the solution in the slope field starting at $(-1, -2)$. Since the solution is linear, it follows the slope lines that have the same slope as the slope line at $(-1, -2)$. At $x = 3$ the solution curve would be at $y = 2$.

8 If $f(x) = \sin(e^{3x})$, then $f'(x) =$

(单选题)

A $\cdot\cos(e^{3x})$

B $\cdot\cos(3e^{3x})$

C $\cdot e^{3x}\cos(e^{3x})$

D $\cdot 3e^{3x}\cos(e^{3x})$

正确答案：D

解析

Answer D

Correct. The chain rule must be used twice in doing the differentiation, as follows.

$f'(x) = \cos(e^{3x}) \cdot \frac{d}{dx}(e^{3x}) = \cos(e^{3x}) \cdot e^{3x} \cdot \frac{d}{dx}(3x) = \cos(e^{3x}) \cdot e^{3x} \cdot 3$

9 If the series $\sum_{n=0}^{\infty} (-1)^n \frac{B^{n+1}}{A^n}$ converges, where $A \neq 0$ and $B \neq 0$, what is the sum of the series?

(单选题)

A . $\frac{AB}{A-B}$

B . $\frac{AB}{A+B}$

C . $\frac{A}{A-B}$

D . $\frac{A}{A+B}$

正确答案：B

解析

Answer B

Correct. This is a geometric series with ratio $r = -\frac{B}{A}$ and first term B. The series converges to $\frac{B}{1-\left(-\frac{B}{A}\right)} = \frac{B}{1+\frac{B}{A}} = \frac{B}{\frac{A+B}{A}} = \frac{AB}{A+B}$.

10 Of the following series, which can be used with the comparison test to show that $\sum_{n=3}^{\infty} \frac{\ln n}{n}$ diverges?

(单选题)

A . $\sum_{n=3}^{\infty} \frac{1}{n}$

B . $\sum_{n=3}^{\infty} \frac{1}{n\sqrt{n}}$

C . $\sum_{n=3}^{\infty} \frac{1}{n^2}$

D . $\sum_{n=3}^{\infty} \ln n$

正确答案：A

解析

Answer A

Correct. Since $\ln n > 1$, for $n \geq 3$, $\frac{1}{n} < \frac{\ln n}{n}$ for $n \geq 3$. Because the series $\sum_{n=3}^{\infty} \frac{1}{n}$ diverges, the comparison test can therefore be used to show that $\sum_{n=3}^{\infty} \frac{\ln n}{n}$ also diverges.

11 A particle moves in the xy-plane so that its position at any time t is given by the parametric equations $x(t) = t^2 - t^3$ and $y(t) = 9 - 6t$. What is the speed of the particle at time $t = 2$?

(单选题)

A . $\frac{3}{4}$

B .5

C .10

D .14

正确答案：C

解析

Answer C

Correct. The speed of the particle at time t is $\sqrt{(x'(t))^2 + (y'(t))^2}$. $x'(t) = 2t - 3t^2 \Rightarrow x'(2) = 4 - 12 = -8$ $y'(t) = -6 \Rightarrow y'(2) = -6$ The speed at time $t = 2$ is therefore $\sqrt{(-8)^2 + (-6)^2} = \sqrt{64 + 36} = \sqrt{100} = 10$.

12 Which of the following is the solution to the differential equation $\frac{dy}{dx} = -3y$ with the initial condition $y(0) = 7$?

(单选题)

A. $y = 7e^{-3x}$

B. $y = e^{-3x} + 6$

C. $y = -\frac{3}{2}x^2 + 7$

D. $y = \sqrt{49 - 6x}$

正确答案：A

解析

Answer A

Correct. The exponential growth and decay model, $\frac{dy}{dx} = ky$, with initial condition $y = y_0$ when $x = 0$, has solutions of the form $y = y_0e^{kx}$. Here, $k = -3$ and $y_0 = 7$. Therefore, the solution is $y = 7e^{-3x}$. The differential equation can also be solved by using separation of variables and the initial condition to determine the appropriate value for the arbitrary constant. $\frac{dy}{dx} = -3y \Rightarrow \frac{1}{y}\,dy = -3\,dx$

$\int \frac{1}{y}\,dy = \int -3\,dx \Rightarrow \ln y = -3x + C \Rightarrow y = e^{-3x+C} = e^C e^{-3x} = ke^{-3x}$ $7 = ke^0 = k$ $y = 7e^{-3x}$ Note: Absolute value was not used in the antiderivative of $\frac{1}{y}$, since the initial value of y is positive.

13 Consider the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{\cos\left(\frac{1}{n}\right)}$. Which of the following statements is true?

(单选题)

A .The series converges absolutely.

B .The series converges conditionally.

C .The series diverges.

D .It cannot be determined whether the series converges or diverges from the information given.

正确答案：C

解析

Answer C

Correct. The series diverges by the n*th* term test. As $n \to \infty$, $\cos\left(\frac{1}{n}\right)$ approaches cos(0) = 1. Therefore, as $n \to \infty$, the terms $\frac{(-1)^n}{\cos\left(\frac{1}{n}\right)}$ are alternating between values approaching -1 and 1. This shows that $\lim_{n \to \infty} \frac{(-1)^n}{\cos\left(\frac{1}{n}\right)}$ does not exist. Therefore, since the terms do not approach 0 in the limit as $n \to \infty$, the series does not converge.

14 If f is a function such that $\lim_{x \to \infty} f(x) = 0$, which of the following could be an expression for $f(x)$?

(单选题)

A . $\frac{x^2}{x^2+4}$

B . $\frac{\sin x}{x}$

C . $\cos x$

D . e^x

正确答案：B

解析

Answer B

Correct. Because $-\frac{1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x}$ for $x > 0$ and $\lim_{x \to \infty}\left(-\frac{1}{x}\right) = \lim_{x \to \infty} \frac{1}{x} = 0$, it follows that $\lim_{x \to \infty} \frac{\sin x}{x} = 0$ by the squeeze theorem.

15 $\int \frac{x+1}{x(x+3)}\,dx =$

(单选题)

A . $\frac{1}{3}\ln|x| + \frac{2}{3}\ln|x+3| + C$

B . $\frac{1}{3}\ln|x| + \frac{2}{9}\ln|x+3| + C$

C . $\frac{2}{3}\ln|x| + \frac{1}{3}\ln|x+3| + C$

D . $\frac{2}{3}\ln|x| + \frac{1}{9}\ln|x+3| + C$

正确答案：A

解析

Answer A

Correct. Evaluating this integral can be done by using the method of partial fractions.
$\frac{x+1}{x(x+3)} = \frac{A}{x} + \frac{B}{x+3} = \frac{A(x+3)+Bx}{x(x+3)} = \frac{(A+B)x+3A}{x(x+3)}$ Equating terms in the numerator,
$\begin{matrix} A+B=1 \\ 3A=1 \end{matrix} \Rightarrow A = \frac{1}{3}, B = \frac{2}{3}$. Now evaluate the integral using the partial fractions.
$\int \frac{x+1}{x(x+3)}\,dx = \int (\frac{1}{3}\cdot\frac{1}{x} + \frac{2}{3}\cdot\frac{1}{x+3})\,dx = \frac{1}{3}\ln|x| + \frac{2}{3}\ln|x+3| + C$

16 $f(x) = \begin{cases} x^2 - 2x & \text{for } x \le 4 \\ g(x) & \text{for } x > 4 \end{cases}$

Let f be the function defined as shown. If f is differentiable, which of the following could define $g(x)$?

(单选题)

A .$6x$

B . $x\frac{3}{2}$

C .$x^2 - 8$

D .$\frac{1}{8}x^3$

正确答案：D

解析

Answer D

Correct. For f to be differentiable at $x = 4$, two conditions must hold for a differentiable function $g(x)$. (1) $g(4) = (x^2 - 2x)|_{x=4} = 16 - 8 = 8$ (continuity at $x = 4$) (2) $g'(4) = \frac{d}{dx}(x^2 - 2x)|_{x=4} = (2x - 2)|_{x=4} = 6$ (differentiability at $x = 4$) The differentiable function $g(x) = \frac{1}{8}x^3$ satisfies both conditions. (For more information, see "Differentiability of Piecewise Defined Functions," https://apcentral.collegeboard.org/courses/resources/differentiability-piecewise-defined-functions, in the Classroom Resources section of the AB/BC courses at AP Central.)

17 The function $y = f(x)$ satisfies the differential equation $\frac{dy}{dx} = 2xy$ with the initial condition $f(0) = 5$. What is the value of $f(2)$?

(单选题)

A . $\frac{5}{21}$

B . $\sqrt{33}$

C . $4 + e^4$

D . $5e^4$

正确答案：D

解析

Answer D

Correct. The differential equation can be solved using separation of variables and the initial condition to determine the appropriate value for the arbitrary constant. $\frac{dy}{dx} = 2xy \Rightarrow \frac{1}{y}\,dy = 2x\,dx$ $\int \frac{1}{y}\,dy = \int 2x\,dx \Rightarrow \ln y = x^2 + C \Rightarrow y = e^{x^2+C} = e^C e^{x^2} = Ke^{x^2}$ $5 = Ke^0 = K$ $y = 5e^{x^2} \Rightarrow f(2) = 5e^4$ Note: Absolute value was not used in the antiderivative of $\frac{1}{y}$ since the initial value of y is positive.

18 Which of the following gives the length of the curve $y = \tan x$ from $x = -\frac{\pi}{4}$ to $x = \frac{\pi}{4}$?

(单选题)

A . $\int_{\frac{\pi}{4}}^{\frac{\pi}{4}} \sqrt{1 + \tan x}\,dx$

B .

$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sqrt{1+\tan^2 x}\, dx$

C. $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sqrt{1+\sec^2 x}\, dx$

D. $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sqrt{1+\sec^4 x}\, dx$

正确答案：D

解析

Answer D

Correct. The length of the graph of $y = f(x)$ from $x = a$ to $x = b$ is given by $\int_a^b \sqrt{1+(f'(x))^2}\, dx$. If $f(x) = \tan x$, then $f'(x) = \sec^2 x$. Therefore the length of the curve from $x = -\frac{\pi}{4}$ to $x = \frac{\pi}{4}$ is $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sqrt{1+(\sec^2 x)^2}\, dx = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sqrt{1+\sec^4 x}\, dx$.

19 If $y = f(x)$ is a function such that $\frac{dy}{dx} = \frac{2x-y}{x}$, then $\frac{d^2y}{dx^2} =$

(单选题)

A. $\frac{2y-2x}{x^2}$

B. $\frac{y-xy}{x^2}$

C . $\frac{y-x}{x^2}$

D . $\frac{y}{x^2}$

正确答案：A

解析

Answer A

Correct. The second derivative can be computed using the quotient rule and implicit differentiation. To simplify using the quotient rule, first rewrite $\frac{dy}{dx}$ as $2-\frac{y}{x}$. $\frac{d^2y}{dx^2}=\frac{d}{dx}\left(\frac{dy}{dx}\right)=\frac{d}{dx}\left(2-\frac{y}{x}\right)=-\frac{\frac{dy}{dx}\cdot x-y\cdot 1}{x^2}=\frac{y-(2x-y)}{x^2}=\frac{2y-2x}{x^2}$

20 Which of the following series converges to $\frac{1}{e}$?

(单选题)

A . $\sum_{n=0}^{\infty}\frac{1}{n!}$

B . $\sum_{n=0}^{\infty}\frac{(-1)^n}{n!}$

C . $\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n)!}$

D . $\sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{(2n+1)!}$

正确答案：B

解析

Answer B

Correct. The series $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ is the Maclaurin series for e^x and converges to e^x for all x. Letting $x = -1$ gives $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!} = e^{-1} = \frac{1}{e}$.

21 The function g has a continuous second derivative. If $g(0) = 2$, $g'(0) = 4$, $g(3) = 5$, and $g'(3) = 6$, what is the value of $\int_0^3 x \cdot g''(x)\,dx$?

(单选题)

A .9

B .15

C .21

D .27

正确答案：B

解析

Answer B

Correct. The integral can be evaluated using integration by parts, where $\int u\,dv = uv - \int v\,du$, as follows.
$u = x \Rightarrow du = dx \quad dv = g''(x)\,dx \Rightarrow v = g'(x)$
$\int_0^3 xg''(x)\,dx = xg'(x)\Big|_0^3 - \int_0^3 g'(x)\,dx = (3g'(3) - 0) - (g(3) - g(0)) = 18 - (5 - 2) = 15$ The Fundamental Theorem of Calculus was used to evaluate the second definite integral.

22 The Maclaurin series for the function f is given by $\sum_{n=1}^{\infty} \frac{(-2)^n x^{2n}}{n}$. What is the value of $f^{(4)}(0)$?

(单选题)

A .2

B .4

C .8

D .48

正确答案：D

解析

Answer D

Correct. The coefficient of the x^{2n} term in the Maclaurin series for f is $\frac{f^{(2n)}(0)}{(2n)!}$. The x^4 term corresponds to $n = 2$. Therefore, $\frac{(-2)^2}{2} = \frac{f^{(4)}(0)}{4!} \Rightarrow f^{(4)}(0) = \frac{(-2)^2}{2} \cdot 4! = 2 \cdot 24 = 48$.

23 Let f be a differentiable function such that $f(\pi) = 2$ and $f'(\pi) = -3$. What is the slope of the line tangent to the polar curve $r = f(\theta)$ at $\theta = \pi$?

(单选题)

A .-3

B .-2

C .$-\frac{3}{2}$

D .$-\frac{2}{3}$

正确答案：D

解析

Answer D

Correct. The slope of the polar curve is $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$.

$x = r\cos\theta = f(\theta)\cos\theta \Rightarrow \frac{dx}{d\theta}\Big|_{\theta=\pi} = f'(\pi)\cos\pi - f(\pi)\sin\pi = (-3)(-1) - (2)(0) = 3$

$y = r\sin\theta = f(\theta)\sin\theta \Rightarrow \frac{dy}{d\theta}\Big|_{\theta=\pi} = f'(\pi)\sin\pi + f(\pi)\cos\pi = (-3)(0) + (2)(-1) = -2$

$$\frac{dy}{dx}\Big|_{\theta=\pi} = \frac{\frac{dy}{d\theta}\Big|_{\theta=\pi}}{\frac{dx}{d\theta}\Big|_{\theta=\pi}} = \frac{-2}{3} = -\frac{2}{3}$$

24 $\int_{-\infty}^{1} x^2 2^{x^3}\, dx$ is

(单选题)

A . $\frac{6}{\ln 2}$

B . $\frac{2}{3\ln 2}$

C . $\frac{2\ln 2}{3}$

D .divergent

正确答案：B

解析

Answer B

Correct. This is an improper integral since the region over which the integrand is being integrated is unbounded. $\int_{-\infty}^{1} x^2 2^{x^3}\,dx = \lim_{t\to-\infty} \int_{t}^{1} x^2 2^{x^3}\,dx$ The integral can be evaluated by using substitution of variables with $u = x^3$. $u = x^3 \Rightarrow \frac{du}{dx} = 3x^2 \Rightarrow dx = \frac{1}{3x^2}\,du$ When $x = 1$, $u = 1$. When $x = t$, $u = t^3$. Substituting for x^3, for dx, and for the limits of integration, $\lim_{t\to-\infty} \int_{t}^{1} x^2 2^{x^3}\,dx = \lim_{t\to-\infty} \int_{t^3}^{1} \frac{2^u}{3}\,du = \lim_{t\to-\infty} \left.\frac{2^u}{3\ln 2}\right|_{u=t^3}^{u=1} = \lim_{t\to-\infty} \left(\frac{2}{3\ln 2} - \frac{2^{t^3}}{3\ln 2}\right) = \frac{2}{3\ln 2}$, since $\lim_{t\to-\infty} 2^{t^3} = 0$.

25 $\int \frac{dx}{x^2+4x+5} =$

(单选题)

A. $\tan^{-1}(x+2)+C$

B. $\ln\left|x^2+4x+5\right|+C$

C. $\frac{1}{2x+4}\ln\left|x^2+4x+5\right|+C$

D. $\frac{1}{\frac{1}{3}x^3+2x^2+5x}+C$

正确答案：A

解析

Answer A

Correct. The integral can be computed by completing the square in the denominator.

$$\int \frac{dx}{x^2+4x+5} = \int \frac{dx}{x^2+4x+4+1} = \int \frac{1}{(x+2)^2+1}\,dx = \tan^{-1}(x+2) + C$$

26 Let f be the function defined by $f(x) = x^3 + x$. If g is the inverse function of f, what is the slope of the line tangent to the graph of g at the point (2, 1) ?

(单选题)

A . $-\frac{1}{4}$

B . $-\frac{1}{13}$

C . $\frac{1}{13}$

D . $\frac{1}{4}$

正确答案：D

解析

Answer D

Correct. The slope of the line tangent to the graph of g at the point (2, 1) is $g'(2)$. Since g is the inverse of f, $f(g(x)) = x$ for all x. By the chain rule, $f'(g(x))g'(x) = 1$, where $f'(x) = 3x^2 + 1$. Therefore, g'.

27

x	0	2	3	5
$f(x)$	2	3	6	12

The function f is continuous on the closed interval $[0,5]$ and differentiable on the open interval $(0,5)$. Selected values of f are given in the table shown. The value $c = 4$ satisfies the conclusion of the Mean Value Theorem for f on the closed interval $[0,5]$. What is the slope of the line tangent to the graph of f at $x = 4$?

(单选题)

A .2

B .2.8

C .4.5

D .10

正确答案：A

解析

Answer A

Correct. Since value $c = 4$ satisfies the conclusion of the Mean Value Theorem for f on the closed interval $[0,5]$, $f'(4) = f'(c) = \frac{f(5)-f(0)}{5-0} = \frac{12-2}{5} = \frac{10}{5} = 2$.

28 Let f be the function with $f(0) = 0$ and derivative given by $f'(x) = \cos\left(x^2\right)$. Which of the following is the Maclaurin series for f ?

(单选题)

A . $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$

B . $\sum_{n=0}^{\infty} \frac{(-1)^n x^{4n}}{(2n)!}$

C . $\sum_{n=0}^{\infty} \frac{(-1)^n x^{4n+1}}{(4n+1)(2n)!}$

D . $\sum_{n=0}^{\infty} \frac{(-1)^n (4n) x^{4n-1}}{(2n)!}$

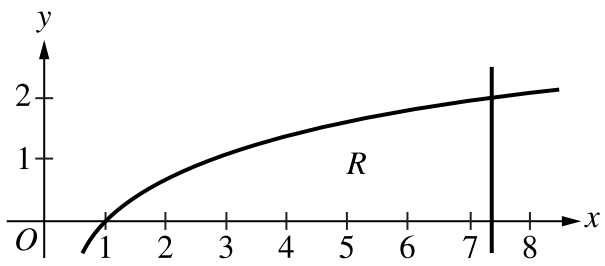
正确答案：C

解析

Answer C

Correct. The Maclaurin series for $\cos x$ is $\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$. Substituting x^2 for x in the Maclaurin series for $\cos x$ gives $f'(x) = \cos\left(x^2\right) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{4n}}{(2n)!}$. Term-by-term integration then gives $f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{4n+1}}{(4n+1)(2n)!} + C$. Since $f(0) = 0$, it follows that $C = 0$.

29



Let R be the region bounded by the graph of $y = \ln x$, the vertical line $x = e^2$, and the x-axis, as shown in the figure.

Which of the following integrals are equal to the area of R ?

1. $\int_1^{e^2} \ln x \, dx$
2. $\int_0^2 e^y \, dy$
3. $\int_0^2 \left(e^2 - e^y\right) dy$

(单选题)

A .I only

B .III only

C .I and II only

D .I and III only

正确答案：D

解析

Answer D

Correct. If the region is sliced along the x-axis so that the area is expressed as a definite integral in terms of x, then the top curve is $y = \ln x$ and the bottom curve is $y = 0$, the x-axis, for $1 \le x \le e^2$. Therefore the area would be given by $\int_1^{e^2} (\ln x - 0)\, dx = \int_1^{e^2} \ln x\, dx$. The region could also be sliced along the y-axis so that the area is expressed as a definite integral in terms of y. Then the right curve is the vertical line $x = e^2$ and the left curve is $x = e^y$ for $0 \le y \le 2$. In this case, the area would be given by $\int_0^2 \left(e^2 - e^y\right) dy$. The integral $\int_0^2 e^y\, dy$ would give the area of the region bounded by the graph of $y = \ln x$, the y-axis, and the horizontal line $y = 2$. This would be $2e^2$ minus the area of R.

30 $\lim_{n \to \infty} \sum_{k=1}^{n} \left(\frac{3}{n}k\right)^2 \left(\frac{3}{n}\right)$ is

(单选题)

A .1

B .3

C .9

D .nonexistent

正确答案：C

解析

Answer C

Correct. This is the limit of a right Riemann sum with n terms built from rectangles of width $\Delta x = \frac{3}{n}$ over an interval of width 3. The first right endpoint is $\frac{3}{n}$ and the last right endpoint is $\frac{3n}{n} = 3$. This means that the integration can be taken over the interval $[0, 3]$ with the height of each rectangle determined by the function $f(x) = x^2$ evaluated at the right endpoints $x_k = \frac{3}{n}k = k\Delta x$ for k from 1 to n. The limit of the Riemann sum will be $\lim_{n \to \infty} \sum_{k=1}^{n} f(x_k)\Delta x = \int_0^3 f(x)\,dx = \int_0^3 x^2\,dx = \frac{x^3}{3}\Big|_0^3 = 9.$

Section I, Part B

76 If $f'(x) = 1 + 2x - x^3 - \sqrt{x^4 + 1}$, then f has a local minimum at $x =$

(单选题)

A .-1.836

B .-1.117

C .0

D .1.181

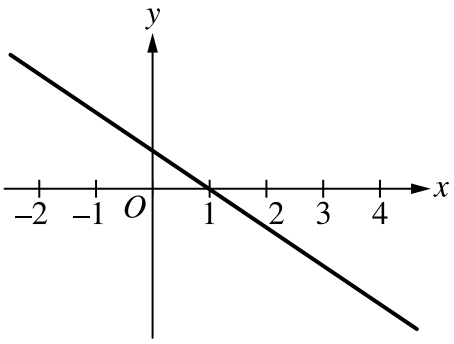
正确答案：C

解析

Answer C

Correct. The function f will have a local minimum at a critical point where the graph of f' changes from negative to positive. Graphing f' on the calculator reveals only one critical point where f' changes from negative to positive, which is at $x = 0$.

77

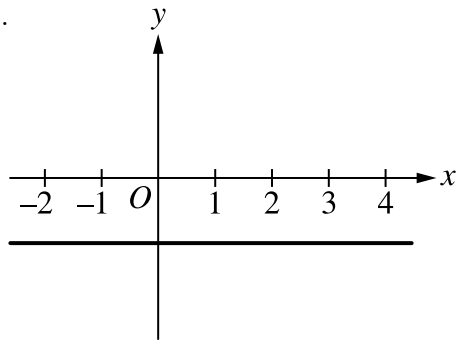


Graph of g

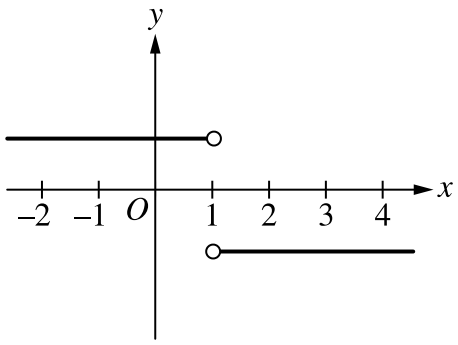
The preceding figure shows the graph of g. If $g(x) = \int_1^x f(t)\,dt$, which of the following could be the graph of $y = f(x)$?

(单选题)

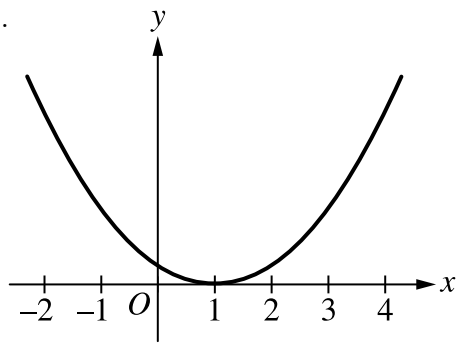
A .



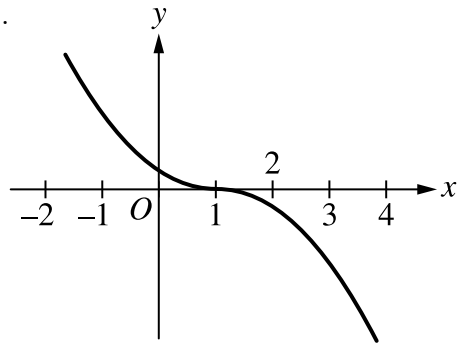
B .



C .



D .



正确答案：A

解析

Answer A

Correct. Because g is decreasing on the interval $[-2, 4]$, $f(x) = g'(x) < 0$ for $-2 < x < 4$. In addition, the graph of g suggests that g is a linear function. Therefore, $f(x) = g'(x)$ would be constant for all x. This is the only graph that satisfies those conditions.

78

x	-1	1
$f(x)$	10	0
$f'(x)$	-1	-9
$f''(x)$	23	-13
$f'''(x)$	-42	6

The function f has derivatives of all orders for all real numbers. The table shown gives values of f and its first three derivatives at selected values of x. What is the value of $\int_{-1}^{1} f''(x)\,dx$?

(单选题)

A .-36

B .-10

C .-8

D .48

正确答案：C

解析

Answer C

Correct. By the Fundamental Theorem of Calculus, $\int_{-1}^{1} f''(x)\,dx = f'(1) - f'(-1) = -9 - (-1) = -8$.

79 On a certain day, an industrial plant manufactures steel at a rate modeled by $R(t) = 10 + 30e^{-0.6t}$ tons per hour, where t is measured in hours since the plant opened that day. What is the average rate at which the plant manufactures steel, in tons per hour, over the time interval $0 \le t \le 5$?

(单选题)

A .11.494

B .19.502

C .25.747

D .97.511

正确答案：B

解析

Answer B

Correct. The average rate at which the plant manufactures steel over the time interval $0 \le t \le 5$ is given by the average value of R over this interval. The average value of a function f on an interval $a \le t \le b$ is given by $\frac{1}{b-a}\int_a^b f(t)\,dt$. $\frac{1}{5}\int_0^5 R(t)\,dt = \frac{1}{5}\int_0^5 \left(10 + 30e^{-0.6t}\right)dt = 19.052$ The calculator is used to evaluate the definite integral.

80 The power series $\sum_{n=0}^{\infty} a_n (x+1)^n$ converges at $x = 3$. Which of the following must be true?

（单选题）

A . $\sum_{n=0}^{\infty} a_n (x+1)^n$ converges at $x=-5$.

B . $\sum_{n=0}^{\infty} a_n (x+1)^n$ diverges at $x=-5$.

C . $\sum_{n=0}^{\infty} a_n (x+1)^n$ converges at $x=-4$.

D . $\sum_{n=0}^{\infty} a_n (x+1)^n$ diverges at $x=4$.

正确答案：C

解析

Answer C

Correct. The radius of convergence is at least 4, since the center is at $x=-1$ and the series converges at $x=3$. Therefore, the interval of convergence includes the open interval $(-5,3)$. Since -4 is in this interval, the series must converge at $x=-4$.

81 A thin rectangular sheet of metal expands in a hot oven. At the instant when the width of the rectangular sheet of metal is 200 millimeters and the length is 300 millimeters, the width is increasing at a rate of 2 millimeters per minute and the length is increasing at a rate of 3 millimeters per minute. What is the rate of change of the area, in square millimeters per minute, of the rectangular sheet of metal at that instant?

(单选题)

A .5

B .6

C .600

D .1200

正确答案：D

解析

Answer D

Correct. The area of the sheet is given by $A = wh$, where both the width w and the length h are functions of time. At the instant described, $w = 200$ and $\frac{dw}{dt} = 2$, while $h = 300$ and $\frac{dh}{dt} = 3$. The product rule and chain rule can be used to determine that at this instant $\frac{dA}{dt} = \frac{dw}{dt} \cdot h + w \cdot \frac{dh}{dt} = (2)(300) + (200)(3) = 600 + 600 = 1200$.

82 A particle travels along the x-axis so that its velocity at time $t \geq 0$ is given by $\frac{dx}{dt} = \sin\left(t^2\right)$. What is the total distance the particle travels from $t = 0$ to $t = 2.7$?

(单选题)

A .0.519

B .0.845

C .1.448

D .2.887

正确答案：C

解析

Answer C

Correct. The definite integral of speed represents the particle's total distance traveled over an interval of time. $\int_0^{2.7} |v(t)| \, dt = \int_0^{2.7} \left|\sin\left(t^2\right)\right| \, dt = 1.448$ The definite integral is evaluated on the calculator.

83 The function f is continuous on the closed interval $[a, b]$, where a and b are constants and $a < b$. If $f(a) < 0 < f(b)$, which of the following statements must be true?

(单选题)

A .There is at least one value of x in the interval (a, b) such that $f(x) = 0$.

B .There is at least one value of x in the interval (a, b) such that $f'(x) = \frac{f(b) - f(a)}{b - a}$.

C .$f'(x) > 0$ for all x in the interval (a, b).

D .$f(a) < f(x) < f(b)$ for all x in the interval (a, b).

正确答案：A

解析

Answer A

Correct. This statement is true by the Intermediate Value Theorem, since f is continuous on the closed interval $[a, b]$ and $f(a) < 0 < f(b)$.

84 The alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2n+2}$ converges to S. If the sum of the first six terms of the series is used to approximate S, what is the alternating series error bound?

(单选题)

A .$\frac{1}{6}$

B .$\frac{1}{7}$

C .$\frac{1}{14}$

D .$\frac{1}{16}$

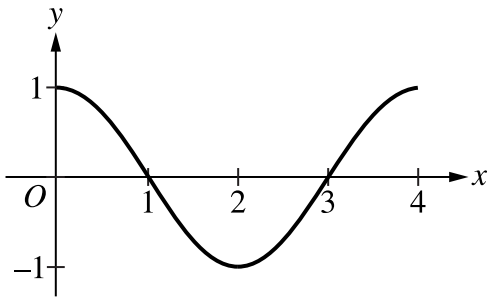
正确答案：D

解析

Answer D

Correct. The alternating series error bound is the absolute value of the first omitted term in the series. Since the first six terms of the series are used to approximate S, the first omitted term corresponds to $n = 7$. Therefore, the alternating series error bound is $\left|S - \sum_{n=1}^{6} \frac{(-1)^{n+1}}{2n+2}\right| \leq \left|\frac{(-1)^8}{2(7)+2}\right| = \frac{1}{16}$.

85



Graph of f

The graph of a twice-differentiable function f is shown for $0 < x < 4$. The graph of f has a horizontal tangent at $x = 2$ and points of inflection at $x = 1$ and $x = 3$. On what intervals do $f'(x)$ and $f''(x)$ have the same sign?

(单选题)

A .$0 < x < 1$ only

B .$2 < x < 4$

C .$0 < x < 1$ and $2 < x < 3$

D .$1 < x < 2$ and $3 < x < 4$

正确答案：C

解析

Answer C

Correct. $f'(x)$ and $f''(x)$ are both negative where the graph of f is decreasing and concave down. This occurs on the interval $0 < x < 1$. $f'(x)$ and $f''(x)$ are both positive where the graph of f is increasing and concave up. This occurs on the interval $2 < x < 3$.

86

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
2	-6	7	4	-1
4	5	6	0	5

The table shown gives values of the twice-differentiable functions f and g and their derivatives f' and g' at selected values of x. What is $\lim_{x \to 2} \frac{g(x^2)}{3x + f(x)}$?

(单选题)

A .$\frac{1}{2}$

B .$\frac{20}{13}$

C .2

D .The limit does not exist.

正确答案：C

解析

Answer C

Correct. Because f and g are differentiable, they are also continuous. Therefore, $\lim_{x \to 2} g\left(x^2\right) = g(4) = 0$ and $\lim_{x \to 2} \left(3x + f(x)\right) = 6 + f(2) = 0$. The limit is of indeterminate form and can be evaluated using L'Hospital's Rule. Because f' and g' are differentiable, they are also continuous. Therefore,

$\lim_{x \to 2} \frac{g\left(x^2\right)}{3x + f(x)} = \lim_{x \to 2} \frac{2xg'\left(x^2\right)}{3 + f'(x)} = \frac{4g'(4)}{3 + f'(2)} = \frac{4(5)}{3 + 7} = \frac{20}{10} = 2.$

87 The velocity of a particle moving along the x-axis is given by $v(t) = \frac{1 - t^3}{1 + t}$ for time $t \geq 0$. Which of the following statements is true about the particle at time $t = 1.5$?

(单选题)

A .The acceleration is negative, and the speed is decreasing.

B .The acceleration is negative, and the speed is increasing.

C .The acceleration is positive, and the speed is decreasing.

D .The acceleration is positive, and the speed is increasing.

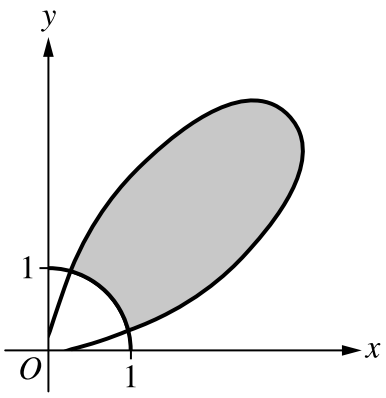
正确答案：B

解析

Answer B

Correct. At time $t = 1.5$, $v(1.5)$ = -0.95 and $a(1.5) = v'(1.5) = -2.32$. Therefore, the acceleration is negative, and the speed is increasing because the velocity and acceleration have the same sign. An alternative solution is to graph the velocity function on the interval $[0, 3]$ and observe that at $t = 1.5$ (in the middle of the window), the graph is below the t-axis and decreasing. This means that the acceleration is negative (since the velocity is decreasing) and that the graph of the speed, the absolute value of the velocity, would be above the t-axis and increasing.

88



The figure shows the graphs of the polar curves $r = 4(\sin(2\theta))^2$ and $r = 1$ for $0 \le \theta \le \frac{\pi}{2}$. What is the area of the shaded region?

(单选题)

A .8.267

B .4.134

C .3.927

D .1.913

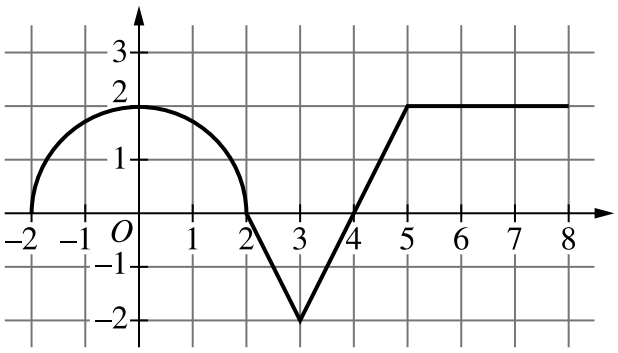
正确答案：B

解析

Answer B

Correct. The area bounded by the two polar curves can be found with a definite integral. Let $r_1 = 1$ be the smaller radius, and let $r_2 = 4\left(\sin(2\theta)\right)^2$ be the larger radius. The two polar curves intersect when $4\left(\sin\left(2\theta\right)\right)^2 = 1$, or $\sin\left(2\theta\right) = \frac{1}{2}$. The two solutions in the interval $[0, 2\pi]$ are $2\theta = \frac{\pi}{6}$ and $2\theta = \frac{5\pi}{6}$, which reduce to $\theta = \frac{\pi}{12}$ and $\theta = \frac{5\pi}{12}$. The area of the shaded region is then $\frac{1}{2}\int_{\frac{\pi}{12}}^{\frac{5\pi}{12}} \left({r_2}^2 - {r_1}^2\right) \mathrm{d}\theta = \frac{1}{2}\int_{\frac{\pi}{12}}^{\frac{5\pi}{12}} \left(16\left(\sin\left(2\theta\right)\right)^4 - 1\right) \mathrm{d}\theta = 4.134$. The definite integral is evaluated with the calculator.

89



Graph of f

The graph of the function f shown consists of a semicircle centered at the origin and three line segments. If $g\left(x\right) = \int_{-2}^{x} \left(f\left(t\right) + 1\right) \mathrm{d}t$, then $g\left(8\right) =$

(单选题)

A .$2\pi + 6$

B .$2\pi + 10$

C .$2\pi + 15$

D .$2\pi + 19$

正确答案：C

解析

Answer C

Correct. $g(8) = \int_{-2}^{8} (f(t) + 1)\, dt$. The definite integral can be evaluated by using properties of integrals as well as geometry and the connection between the definite integral and area.

$g(8) = \int_{-2}^{8} (f(t) + 1)\, dt = \int_{-2}^{8} f(t)\, dt + \int_{-2}^{8} 1\, dt = \left(\frac{1}{2}\pi(2)^2 - \frac{1}{2}(2)(2) + \frac{1}{2}(1)(2) + (3)(2)\right) +$

90 The position of a particle moving in the xy-plane is given by the parametric equations $x(t) = 5\ln(t+5)$ and $y(t) = 5\sin(\pi t)$. At any point on the curve, the slope of the line tangent to the path of the particle is given by $\frac{dy}{dx} = \pi\cos(\pi t)(t+5)$. What is the value of $\frac{d^2y}{dx^2}$ at $t = 1$?

(单选题)

A .-18.850

B .-3.770

C .-3.142

D .0.200

正确答案：B

解析

Answer B

Correct. The chain rule is used to find derivatives of functions defined by parametric equations. $\left.\frac{d^2y}{dx^2}\right|_{t=1} = \left.\frac{d}{dx}\left(\frac{dy}{dx}\right)\right|_{t=1} = \frac{\left.\frac{d}{dt}\left(\frac{dy}{dx}\right)\right|_{t=1}}{\left.\frac{dx}{dt}\right|_{t=1}} = \frac{\left.\frac{d}{dt}\left(\pi\cos(\pi t)(t+5)\right)\right|_{t=1}}{\left.\frac{d}{dt}\left(5\ln(t+5)\right)\right|_{t=1}} = -3.770$ The value of each derivative is computed using the calculator.

Section II, Part A

1 A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work

will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (years)	0	4	9	12
$R(t)$ (bears per year)	150	156	159	161

The rate at which the number of bears in a region is changing is modeled by the differentiable function R, where $R(t)$ is measured in bears per year and t is measured in years. Selected values of $R(t)$ are shown in the table.

(a) Approximate $R'(7)$ using the average rate of change of R over the interval $4 \le t \le 9$. Show the computations that lead to your answer, and indicate units of measure.

(b) Interpret the meaning of $\frac{1}{12}\int_0^{12} R(t)\,dt$ in the context of the problem. Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\frac{1}{12}\int_0^{12} R(t)\,dt$.

(c) The rate at which the number of bears in the region is changing can also be modeled by the function P defined by $P(t) = 150e^{0.02\sqrt{t}}$, where $P(t)$ is measured in bears per year and t is measured in years. There are 300 bears in the region at time $t = 0$. Use this model to find the number of bears in the region at time $t = 12$. Show the setup for your calculations, and give your answer to the nearest whole number of bears.

(d) Using the model defined in part (c), find the time t, for $0 \le t \le 12$, at which the number of bears in the region is a maximum. Justify your answer.

解析

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point a response must provide a numerical difference and a quotient. A response of only or earns the first point.

To earn the second point, a response must have “bear per year per year” (or the equivalent) attached to a numerical value.

The student response accurately includes **both** of the criteria below.

Approximation with supporting work

Units

Solution:

$R'7 \approx \frac{R9 - R4}{9 - 4} = \frac{159 - 156}{5} = 0.6$

$R'7 \approx 0.6$ bear per year per year

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, a response must provide “average,” “rate of change,” and “from time $t = 0$ to $t = 12$ ” (or equivalent). Units are not required to earn the first point.

To earn the second point, at least five of the six factors in the Riemann sum must be correct. The factor of $\frac{1}{12}$ is assessed in the third point.

If there is any error in the Riemann sum, the response does not earn the third point.

A response of only $\frac{1}{12}150 \cdot 4 + 156 \cdot 5 + 159 \cdot 3$ earns the second and third points.

A response that presents the correct average value with accompanying work that does not explicitly present the six factors and/or the sum process does not earn the second point but will earn the third point. For example,

responses of "$\frac{150 \cdot 4 + 156 \cdot 5 + 477}{12}$" or "$600,\ 780,\ 477 \ \rightarrow \ \frac{1857}{12}$" would earn the third point but not the second.

A response that presents an answer of $\frac{156 \cdot 4 + 159 \cdot 5 + 161 \cdot 3}{12}$ (which uses a right Riemann sum) earns 1 of the last 2 points. If this response is simplified, the result must be $\frac{1902}{12}$, $\frac{951}{6}$, $\frac{317}{2}$, or 158.5.

An unsupported response of $\frac{1857}{12}$, $\frac{619}{4}$, or 154.75 does not earn the second or third point.

A response that miscommunicates by equating two unequal quantities, e.g., $\int_0^{12} Rt\ dt = \frac{150 \cdot 4 + 156 \cdot 5 + 159 \cdot 3}{12}$, does not earn the third point.

The student response accurately includes **all three** of the criteria below.

Interpretation

Form of left Riemann sum

Answer

Solution:

$\frac{1}{12} \int_0^{12} Rt\ dt$ is the average rate of change in the number of bears in the region, in bears per year, from time $t = 0$ to time $t = 12$ years.

$$\int_0^{12} Rt\ dt \approx R(0) \cdot (4 - 0) + R(4) \cdot (9 - 4) + R(9) \cdot (12 - 9)$$

$$= 150 \cdot 4 - 0 + 159 \cdot 9 - 4 + 159 \cdot 12 - 9$$

$$= 150 \cdot 4 + 156 \cdot 5 + 159 \cdot 3 = 1857$$

$\frac{1}{12} \int_0^{12} Rt\ dt \approx \frac{1857}{12} = \frac{619}{4} = 154.75$ bears per year

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for a definite integral $\int_a^b Pt\ dt$ with any limits of integration. If the limits are incorrect, the response does not earn the second point.

The second point cannot be earned without the first point, and is earned only for an answer of 2185 or 2185.339 or 2185.338.

Missing differentials:

- A response of only $\int_0^{12} Pt$ or $\int_0^{12} Pt = 1885.3389$ earns the first point but not the second.
- A response of $300 + \int_0^{12} Pt$ earns the first point and in the presence of the correct answer earns the second point.
- A response of $\int_0^{12} Pt + 300$ does not earn the first point but earns the second point for the correct answer.

A response that miscommunicates, e.g., $\int_0^{12} Pt\ dt = 1885.3389 + 300,$ does not earn the second point.

The student response accurately includes **both** of the criteria below.

Integral

Answer

Solution:

$300 + \int_0^{12} Pt\ dt = 300 + 1885.3389 = 2185.3389$

At $t = 12$ years, there are 2185 bears in the region.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that considers $P = 0$ earns the first point.

A response of "at time $t = 12$ " with no justification does not earn the second point.

The second point cannot be earned for any answer other than time $t = 12$.

Alternate solution using Candidates Test:

$Pt = 0$ has no solutions in the interval $0 \le t \le 12,$ so the maximum number of bears occurs at an endpoint of the interval.

The region has 300 bears at time $t = 0$.

From part (c), the region has 2185 bears at time $t = 12$.

For $0 \le t \le 12,$ the maximum number of bears is at time $t = 12$ years.

- The scoring is the same as the model solution.
- A response using this argument may import an incorrect value for the number of bears in the region in year 12 from part (c), provided that value is larger than 300.
- A response using this argument need not report the number of bears at time $t = 12$. Stating that this number is greater than 300 is sufficient.

The student response accurately includes **both** of the criteria below.

Considers sign of P

Answer with justification

Solution:

$Pt > 0$ for all $t \geq 0,$ so the bear population is always increasing.

For $0 \leq t \leq 12,$ the maximum number of bears is at time $t = 12$ years.

2 A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work

will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

For time $t \geq 0$, a particle moves in the xy-plane with position $(x(t), y(t))$, where $x(t) = t + \cos t$ and $\frac{dy}{dt} = \sin\left(t^2\right)$. At time $t = 6$, $y(6) = 4$.

(a) Find the speed of the particle at time $t = 2$. Show the setup for your calculations.

(b) Find the total distance traveled by the particle over the time interval $0 \leq t \leq 6$. Show the setup for your calculations.

(c) Write an equation for the line tangent to the path of the particle at time $t = 6$.

(d) Find the position of the particle at time $t = 0$. Show the setup for your calculations.

解析

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, a response must communicate the use of $t = 2$.

- A response that presents a correct expression for speed in terms of with an incorrect or missing answer does not earn either point. For example, a response of just $\text{Speed} = \sqrt{x't^2 + y't^2}$ does not earn either

point.

A response that presents a correct expression for speed in terms of t with a correct answer, and provides correct communication, earns both points. For example, a response of $\text{Speed} = \sqrt{x't^2 + y't^2}$ and therefore the speed is 0.762 earns both points.

A response that presents a correct expression for speed in terms of t with the correct answer, but provides incorrect or incomplete communication, does not earn the first point but earns the second point. For example, a response of $\text{Speed} = \sqrt{x't^2 + y't^2} = 0.762$ does not earn the first point but earns the second point.

An unsupported answer of 0.762 does not earn either point.

The exact answer is $\sqrt{1 - \sin 2^2 + \sin 4^2}$. This earns both points.

A response that presents an incorrect expression for $x't$ is eligible for the first point only.

Degree mode: A response that presents answers obtained by using a calculator in degree mode does not earn the first point it would have otherwise earned. The response is generally eligible for all subsequent points (unless no answer is possible in degree mode or the question is made simpler by using degree mode). In degree mode, $\sqrt{x'2^2 + y'2^2} = 0.967618$ or 1.001822.

The student response accurately includes **both** of the criteria below.

Expression for speed

Answer

Solution:

$\text{Speed} = \sqrt{x'2^2 + y'2^2} = 0.762218$

At time $t = 2,$ the speed of the particle is 0.762.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

- The first point is earned for presenting the correct integrand in a definite integral, with correct or incorrect limits.

- To earn the second point, a response must have earned the first point, must display correct limits, and must present the value 7.638.
- An unsupported answer of 7.638. does not earn either point.
- A response that imports an incorrect expression for $x't$ from part (a) is eligible for the first point only.

Degree mode: In degree mode, $\int_0^6 \sqrt{1-\sin\ t^2+\sin t^{2^2}}\ dt = 5.914519$ or 6.205287.

The student response accurately includes **both** of the criteria below.

Integrand

Answer

Solution:

$\int_0^6 \sqrt{x't^2+y't^2}\ dt = 7.63809$

The total distance traveled by the particle over the time interval $0 \le t \le 6$ is 7.638.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that writes a linear equation using a slope of -0.775 earns the first point.

A response that writes a correct tangent line equation, in any form, earns both points.

- Note: The slope-intercept form is $y = -0.775x + 9.395$ (or $y = -0.775x + 9.394$).

A response that imports an incorrect expression for $x't$ from part (a) is eligible for both points.

A response that presents an exact correct (or consistent) value of $\frac{dy}{dx}_{t=6}$ earns the first point. If the exact value is simplified incorrectly, the second point is not earned.

Degree mode: In degree mode, an equation of the line tangent to the path at time $t = 6$ could be one of the following:

- $y = 4 + 0.656397x - 6.994522$
- $y = 0.656397x - 0.591183$

- $y = 4 + 0.588860x - 6.994522$
- $y = 0.588860x - 0.118794$

The student response accurately includes **both** of the criteria below.

$\frac{dy}{dx}_{t=6}$

Tangent line equation

Solution:

$\frac{dy}{dx}_{t=6} = \frac{dy\ /\ dt}{dx\ /\ dt}_{t=6} = \frac{\sin 36}{1 - \sin\ 6} = -0.775181$

$x6 = 6 + \cos\ 6 = 6.960170$

$y6 = 4$

An equation of the line tangent to the path at time $t = 6$ is $y = 4 - 0.775x - 6.960$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by an integrand of $\sin t^2$ in either an indefinite integral or a definite integral, with correct or incorrect limits. If the correct limits of integration are never shown, the response does not earn the third point.

If the differential is missing:

- $\int_6^0 \sin t^2$ earns the first point and is eligible to earn the second and third points.
- $\int_6^0 \sin t^2 + y6$ does not earn the first point, but does earn the second point and is eligible for the third point in the presence of the correct answer.
- $y6 + \int_6^0 \sin t^2$ earns the first two points and is eligible for the third.

The second point is earned if $y6$ is added to a definite integral.

A response that communicates an attempt to antidifferentiate $y't$ and produces an equation of the form $yt = ft + c$ earns the second point for stating $4 = f6 + c$.

A response that earns the first point but not the second earns the third point with an answer of $1, 3.362$ (or $1, 3.361$).

The third point is earned for the ordered pair $1, 3.362$ (or $1, 3.361$), or for listing both coordinates as long as they are labeled.

A response that provides a false statement such as $y0 = \int_6^0 \sin t^2$ or $\int_6^0 \sin t^2 = 4 - 0.638459$ does not earn the third point.

Degree mode: In degree mode, $x0 = 1$ and $y0 = 2.778357$.

The student response accurately includes **all three** of the criteria below.

Integrand

Uses initial condition

Answer

Solution:

$$y0 = y6 + \int_6^0 \sin t^2 \, dt = 4 - 0.638459 = 3.361541$$

$$x0 = 0 + \cos 0 = 1$$

The position of the particle at time $t = 0$ is $1, 3.362$ (or $1, 3.361$).

Section II, Part B

3 NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work

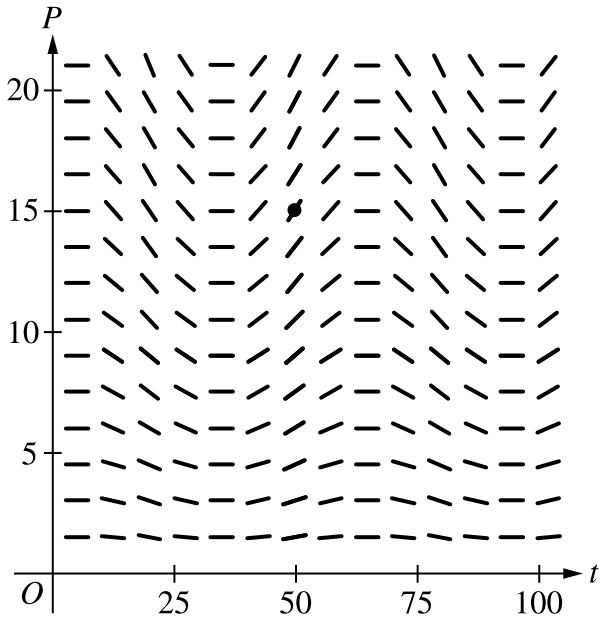
will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Scientists study a population that experiences variations in its rate of growth. The rate of growth of the population is modeled by the differential equation $\frac{dP}{dt} = P \cdot 0.02\cos\left(\frac{\pi}{30}(t+10)\right)$, where t is measured in days since the observation of the population began. Let $P = f(t)$ be the particular solution to the given differential equation satisfying $f(50) = 15$.

(a) A slope field for the given differential equation is shown. Sketch the solution curve that passes through the point $(50, 15)$.



(b) Write an equation for the line tangent to the graph of $P = f(t)$ at time $t = 50$.

(c) It is known that $f'(t) > 0$ and $f''(t) > 0$ on the interval $47 < t < 51$. If the tangent line found in part (b) is used to approximate $f(48)$, is the approximation an underestimate or an overestimate for the actual value of $f(48)$? Give a reason for your answer.

(d) Use separation of variables to find the particular solution $P = f(t)$ to the differential equation $\frac{dP}{dt} = P \cdot 0.02 \cos\left(\frac{\pi}{30}(t + 10)\right)$ that satisfies $f(50) = 15$.

解析

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The solution curve must pass through the point $50, 15,$ extend reasonably close to the left and right edges of the slope field, and have no obvious conflicts with the given line segments.

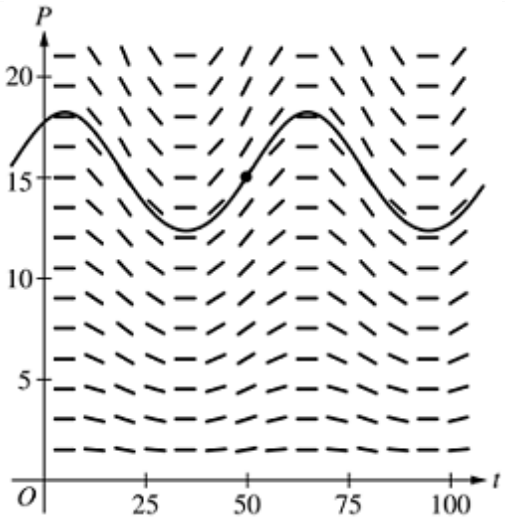
Only portions of the solution curve within the given slope field are considered.

All local maximum/minimum points on the solution curve must occur at horizontal line segments in the slope field.

The student response accurately includes the criteria below.

Solution curve through $50, 15,$

Solution:



Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The tangent line equation can be presented in any equivalent form and can be written in terms of any variables.

The second point can be earned with a tangent line through the point $50, 15$ with a slope equal to the declared value of $\frac{dP}{dt}_{50,\ 15}$.

The student response accurately includes **both** of the criteria below.

Slope at $50, 15$

Tangent line equation

Solution:

$f50 = 15$

$f^{'}50 = \frac{dP}{dt}_{50,\ 15} = 15 \cdot 0.02\cos\frac{\pi}{30} \cdot 60 = 0.3$

An equation for the line tangent to the graph of $P = ft$ at time $t = 50$ is $y = 15 + 0.3t - 50$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The reason must include one of $f''t > 0$, $f't$ is increasing, or ft is concave up.

The point is not earned if the reason includes statements about the behavior of $f't$ or $f''t$ outside the interval $47 < t < 51$.

The student response accurately includes the criteria below.

Underestimate with reason

Solution:

The tangent line approximation of $f48$ is an underestimate because $f''t > 0$ on an interval that contains $t = 48$ and $t = 50$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with no separation of variables earns 0 out of 5 points.

A response that presents an antiderivative of without absolute value symbols is sufficient for the second point.

A response with no constant of integration can earn at most the first 3 points.

A response is eligible for the fourth point only if it has earned the first point and at least 1 of the 2 antiderivative points.

An eligible response earns the fourth point by correctly including the constant of integration in an equation and substituting 50 for *t* and *P*. A response is eligible for the fifth point only if it has earned the first 4 points.

The student response accurately includes **all five** of the criteria below.

Separates variables

Finds one antiderivative

Finds the other antiderivative

Constant of integration and uses initial condition

Solves for *P*

Solution:

$$\int \frac{dP}{P} = \int 0.02\cos\frac{\pi}{30}t + 10\ dt$$

$$\ln P = \frac{0.6}{\pi}\sin\frac{\pi}{30}t + 10 + C$$

$$\ln 15 = \frac{0.6}{\pi}\sin\frac{\pi}{30}50 + 10 + C \Rightarrow C = \ln 15$$

Because $P(50) > 0,\ \ln P = \ln P$.

$$\ln P = \frac{0.6}{\pi}\sin\frac{\pi}{30}t + 10 + \ln 15$$

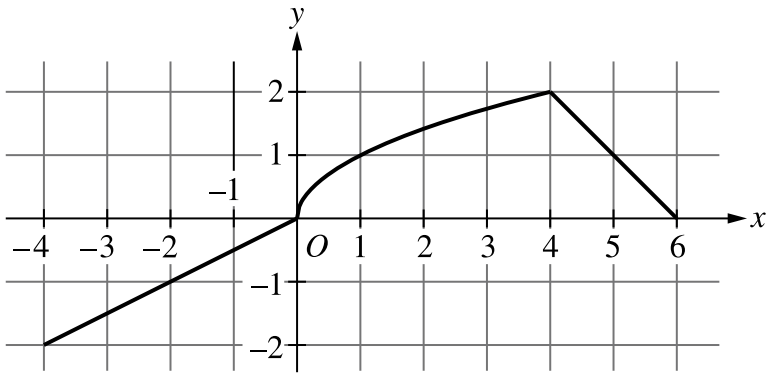
$$P = 15e^{\frac{0.6}{\pi}\sin\frac{\pi}{30}t + 10}$$

4 NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of g

The graph of the function g is shown on the interval $[-4, 6]$. The graph of g consists of two line segments and the portion of the graph of $y = \sqrt{x}$ for $0 \le x \le 4$. Let f be the function defined by $f(x) = g(x) + \int_{-4}^{x} g(t)\,dt$.

(a) Find the value of $f(4)$.

(b) Show that f has a critical point at $x = -1$. Use the Second Derivative Test to determine whether f has a local minimum or a local maximum at $x = -1$. Show the work that leads to your answer.

(c) Find the value of $\lim_{x \to 4^-} \frac{g(x) - g(4)}{x - 4}$.

解析

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response does not need to present $\int_{-4}^{0} g\,t\,dt$ to earn the first point. The value of ± 4 may be determined directly from the graph.

The inclusion of $g4$ is considered in the third point.

The second point is earned by finding the antiderivative $\frac{2}{3}t^{3/2}$ or by determining the area under this portion of the parabola is $\frac{2}{3} \cdot 8$.

A response that finds an antiderivative of the form $k \cdot t^{3/2}$ where $k \neq 0$ and $k \neq \frac{2}{3}$ does not earn the second point but does earn the third point for the consistent answer of $-2 + k \cdot 8$.

The final answer does not need to be simplified. Any error in simplification does not earn the third point.

The student response accurately includes **all three** of the criteria below.

$\pm$ value of $\int_{-4}^{0} gt\ dt$

Antiderivative of $\sqrt{t}$

Answer

Solution:

$$f4 = g4 + \int_{-4}^{4} gt\ dt = 2 + \int_{-4}^{0} gt\ dt + \int_{0}^{4} gt\ dt$$

$$= 2 + -4 + \int_{0}^{4} \sqrt{t}\ dt = -2 + \frac{2}{3}t^{3/2}\Big|_{t=0}^{t=4}$$

$$= -2 + \frac{2}{3} \cdot 8 = \frac{10}{3}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is only earned for the statement $f'x = g'x + gx$ or If there is any equating of unequal quantities, or miscommunication, for example, the response does not earn the first point.

If the first point is not earned, the second point can only be earned for a response that presents

If the first point is earned, a response can earn the second point by stating that

To earn the third point, a response must correctly find or and go on to state that

The fourth point can be earned without any of the first 3 points, provided the response declares a nonzero value or sign of and makes a consistent Second Derivative Test argument based on that value or sign.

The student response accurately includes **all four** of the criteria below.

$f'x = g'x + gx$ OR $f'-1 = g'-1 + g-1$

Shows $f'-1 = 0$

$f''-1$

Answer with reason

Solution:

$f'x = g'x + gx$

$f'-1 = g'-1 + g-1 = \frac{1}{2} + -\frac{1}{2} = 0$

Because $f'-1 = 0$, f has a critical point at $x = -1$.

$f''x = g''x + g'x$

$f''-1 = g''-1 + g'-1 = 0 + \frac{1}{2}$

Because $f'-1 = 0$ and $f''-1 > 0$, the Second Derivative Test indicates that f has a local minimum at $x = -1$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response does not need to state that $\sqrt{x}$ is differentiable.

A minimal response to earn both points is $\frac{1}{2\sqrt{4}}$.

Special case (using L'Hospital's Rule): For $0 \le x \le 4$, $gx = \sqrt{x}$.

$\lim_{x \to 4^-} \sqrt{x} - g4 = 2 - 2 = 0$ and $\lim_{x \to 4^-} x - 4 = 0$.

Because $\sqrt{x}$ is continuous on the interval $0, 4$ and differentiable on the interval $0, 4$, $\lim_{x \to 4^-} gx - g4 = 2 - 2 = 0$, and $\lim_{x \to 4^-} x - 4 = 4 - 4 = 0$, L'Hospital's Rule may be applied:

$$\lim_{x \to 4^-} \frac{gx - g4}{x - 4} = \lim_{x \to 4^-} \frac{g'x}{1} = \lim_{x \to 4^-} \frac{\frac{d}{dx}\sqrt{x}}{1} = \lim_{x \to 4^-} \frac{\frac{1}{2\sqrt{x}}}{1} = \frac{1}{2\sqrt{4}} = \frac{1}{4}.$$

- A response earns the first point by evaluating the limits of the numerator and denominator.
- A response that equates the limit to $\frac{0}{0}$ does not earn the first point but is eligible for the second point. A response may say that the limit is of indeterminate form $\frac{0}{0}$.
- The second point is earned by correctly applying L'Hospital's Rule (by presenting a limit of a ratio of derivatives) and finding the correct answer.

The student response accurately includes **both** of the criteria below.

Uses $gx = \sqrt{x}$

Answer

Solution:

For $0 \leq x \leq 4,\ gx = \sqrt{x}$.

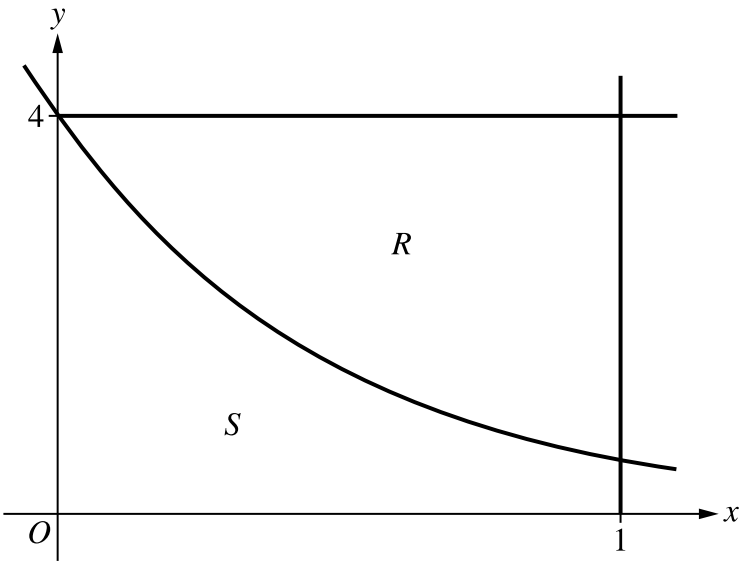
Because $\sqrt{x}$ has a left-hand derivative at $x = 4$, $\lim_{x \to 4^-} \frac{gx - g4}{x - 4} = \frac{d}{dx}\sqrt{x}_{x=4} = \frac{1}{2\sqrt{x}}_{x=4} = \frac{1}{4}$.

5 NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Let f be the function defined by $f\left(x\right) = 4e^{-2x}$. Let R be the region in the first quadrant bounded by the graph of f, the horizontal line $y = 4$, and the vertical line $x = 1$. Let S be the region in the first quadrant bounded by the graph of f, the coordinate axes, and the vertical line $x = 1$. Regions R and S are shown in the figure.

(a) Find the area of region R.

(b) Region S is the base of a solid. For this solid, at each x the cross section perpendicular to the x-axis is a rectangle with area $3x \cdot f\left(x\right)$. Find the volume of the solid.

(c) Write, but do not evaluate, an expression involving an integral that gives the perimeter of region S.

解析

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is also earned for a definite integral with an integrand of $k - fx,\ fx - k,\ k - fx,$ or $fx - k,$ $k \geq 0,$ regardless of any subsequent copy errors. If the limits are incorrect, the response does not earn the third point.

A response that uses an integrand of $fx - 4$:

- can earn the third point if it presents correct communication, such as $\int_0^1 f x - 4\ dx = -2e^{-2} - 2,$ so the area is $2 + 2e^{-2}$.
- does not earn the third point if it includes incorrect communication, such as $\int_0^1 f x - 4\ dx = 2 + 2e^{-2}$.

If a response does not earn the first point, the second point is earned only for determining $\int ae^{-2x}\ dx = -\frac{a}{2}e^{-2x},\ a \neq 0.$

A response earns the third point only if it has earned the first 2 points and presents an answer equivalent to $2 + 2e^{-2}$.

The student response accurately includes **all three** of the criteria below.

$\int_0^1 f x\ dx$

Antiderivative of $4e^{-2x}$

Answer with supporting work

Solution:

$$\text{Area} = \int_0^1 4 - f x\ dx = \int_0^1 4 - 4e^{-2x}\ dx$$

$$= 4x + 2e^{-2x}\Big|_0^1$$

$$= 4 + 2e^{-2} - 0 + 2 = 2 + 2e^{-2}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for $\int_0^1 3x \cdot f x\ dx$ or $c\int_0^1 xe^{-2x}\ dx,$ where $c \neq 0.$ A response using $c \neq 12,$ for example $c = 1,$ will not earn the fourth point.

Incorrect integrals that require integration by parts are eligible for the second and third points. Both of these points are earned by correctly applying integration by parts at least once.

The second point is earned with an implied u and dv in the presence of $-\frac{x}{2}e^{-2x} - \int -\frac{1}{2}e^{-2x}\,dx$. This expression also earns the third point.

The tabular method may be used to show integration by parts.

- In this case, the second point is earned by having evidence to identify $u = x$ and $dv = e^{-2x}$.
- The third point is earned for either $-\frac{x}{2}e^{-2x} - \int -\frac{1}{2}e^{-2x}\,dx$ or $-\frac{x}{2}e^{-2x} - \frac{1}{4}e^{-2x}$.

A response of $\int xe^{-2x}\,dx = -\frac{x}{2}e^{-2x} - \int -\frac{1}{2}e^{-2x}\,dx$ or $\int xe^{-2x}\,dx = -\frac{x}{2}e^{-2x} - \frac{1}{4}e^{-2x}$ (with or without explicit mention of u or dv) earns the second and third points.

The second and third points are unaffected by missing or incorrect limits of integration.

The fourth point is earned only for the correct answer; the value of the volume does not need to be simplified. Any error in simplification does not earn the fourth point.

The student response accurately includes **all four** of the criteria below.

Integral

u and dv

$$\int xe^{-2x}\,dx = -\frac{x}{2}e^{-2x} - \int -\frac{1}{2}e^{-2x}\,dx$$

Answer

Solution:

$$\text{Volume} = \int_0^1 3x \cdot fx\,dx = 12\int_0^1 xe^{-2x}\,dx$$

Using integration by parts,

$$u = x \qquad dv = e^{-2x}\,dx$$

$$du = dx \qquad v = -\frac{1}{2}e^{-2x}$$

$$\int xe^{-2x}\,dx = -\frac{x}{2}\ e^{-2x}\ - \int -\frac{1}{2}e^{-2x}\,dx$$

$$= -\frac{x}{2}\ e^{-2x}\ -\frac{1}{4}\ e^{-2x}\ + C$$

$$\text{Volume} = 12\int_0^1 xe^{-2x}\,dx = 12 - \frac{x}{2}e^{-2x} - \frac{1}{4}e^{-2x}\Big|_0^1$$

$$= 12 - \frac{1}{2}e^{-2} - \frac{1}{4}e^{-2} - 0 - \frac{1}{4} = -9e^{-2} \quad + 3$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for $\int_0^1 \sqrt{1 + f'x^2}\,dx$ or $\int_0^1 \sqrt{1 + -8e^{-2x^2}}\,dx,$ with or without the differential.

The first line of the model solution earns both points.

A response of $\int_0^1 \sqrt{1 + f'x^2} + 5 + 4e^{-2}$ earns the first point but does not earn the second point. (The integrand is ambiguous due to the missing dx.)

The student response accurately includes **both** of the criteria below.

Integral

Answer

Solution:

$$\text{Perimeter} = 4 + 1 + f1 + \int_0^1 \sqrt{1 + f'x^2}\,dx$$

$$= 5 + 4e^{-2} \quad + \int_0^1 \sqrt{1 + -8e^{-2x^2}}\,dx$$

$$= 5 + 4e^{-2} \quad + \int_0^1 \sqrt{1 + 64e^{-4x}}\,dx$$

6 NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

The Taylor series for a function f about $x = 1$ is given by $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n}{4^n} (x-1)^n$ and converges to $f(x)$ on its interval of convergence.

(a) Using the ratio test, find the interval of convergence of the Taylor series for f about $x = 1$. Justify your answer.

(b) Write the first three nonzero terms and the general term of the Taylor series for f', the derivative of f, about $x = 1$.

(c) Let $P_2(x)$ be the second-degree Taylor polynomial for f about $x = 1$. It is known that $0.1 \le \left|f'''(x)\right| \le 0.3$ on the interval $1 \le x \le 1.5$. Use the Lagrange error bound to show that $P_2(1.5)$ differs from $f(1.5)$ by at most $\frac{1}{100}$.

解析

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by presenting a correct ratio with or without absolute values. Once earned, this point cannot be lost. Any errors in simplification or evaluation of the limit will not earn the second point.

A ratio may be expressed in either of the forms $\frac{a_{n+1}}{a_n}$ or $\frac{a_n}{a_{n-1}}$.
An expression equivalent to $\frac{a_n}{a_{n+1}}$ earns the first point and is eligible for the remaining points.

A response that does not present a ratio does not earn the second point.

To earn the second point a response must use the absolute value of the ratio and correctly evaluate the limit of the ratio, including correct limit notation.

A response that does not use absolute value in computing the limit does not earn the second point. However, this response is eligible for the third point if the expression $-1 < \frac{x-1}{4} < 1$ is presented.

A response that presents an incorrect limit of the form $\frac{x-1}{b}$, where $b \neq 0$, is eligible for the third point. In this case the third point would be earned for finding a consistent interior interval of convergence.

To earn the third point, the interval presented must have interior $-3, 5$ or $1-b, 1+b$;
$x - 1 < 4$ is not sufficient to earn the third point.

The fourth point is earned for considering both endpoints of the correct interval, or both endpoints of an incorrect interval that has earned the third point.

To earn the fifth point, the response must correctly analyze the series at $x = -3$ and $x = 5$, and present the correct interval of convergence.

A response that uses the root test is eligible only for the third, fourth, and fifth points.

The student response accurately includes **all five** of the criteria below.

Sets up ratio

Computes limit of ratio

Identifies interior of interval of convergence

Considers both endpoints

Analysis and interval of convergence

Solution:

$$\frac{-1^{n+2}\frac{n+1}{4^{n+1}}x-1^{n+1}}{-1^{n+1}\frac{n}{4^n}x-1^n}=\frac{x-1}{4}\cdot\frac{n+1}{n}$$

$$\lim_{n\to\infty}\frac{x-1}{4}\cdot\frac{n+1}{n}=\frac{x-1}{4}$$

$$\frac{x-1}{4}=\frac{1}{4}x-1<1\ \Rightarrow x-1<4\Rightarrow\ -3<x<5$$

When $x=\ -3$,

the series is $\sum_{n=1}^{\infty}-1^{n+1}\frac{n}{4^n}-4^n=\sum_{n=1}^{\infty}-n=\ -1-2-3-\ \cdots$,

which diverges because $\lim_{n\to\infty}-n\neq 0$.

When $x=5$,

the series is $\sum_{n=1}^{\infty}-1^{n+1}\frac{n}{4^n}4^n=\sum_{n=1}^{\infty}-1^{n+1}n=1-2+3-4+\ \cdots$,

which diverges because $\lim_{n\to\infty}-1^{n+1}n\neq 0$.

The interval of convergence of the Taylor series for f is $-3<x<5$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by presenting the first three nonzero terms in a list or as part of a polynomial or series.

The second point is earned by identifying the general term (either individually or as part of a polynomial or series).

Read “ $=$
” as “ $\approx$
” as necessary.

The student response accurately includes **both** of the criteria below.

First three terms

General term

Solution:

The Taylor series for f about $x = 1$
is $\frac{1}{4}x - 1 - \frac{2}{16}x - 1^2 + \frac{3}{64}x - 1^3 + \cdots + -1^{n+1}\frac{n}{4^n}x - 1^n + \cdots$.

Differentiating term-by-term, the Taylor series for f'
about $x = 1$
is $\frac{1}{4} - \frac{4}{16}x - 1 + \frac{9}{64}x - 1^2 + \cdots + (-1)^{n+1}\frac{n^2}{4^n}(x-1)^{n-1} + \cdots$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for presenting either $\frac{\max\limits_{1 \le x \le 1.5} f'''x}{3!}1.5 - 1^3$
or $\frac{0.3}{6}\frac{1}{2}^3$
with no mistake in arriving at this bound.

If there is miscommunication or notational mistakes in arriving at the error bound, the response earns the first point but is not eligible for the second point.

To earn the second point, a response must have earned the first point and present the inequality
$\text{Error} \le \frac{1}{160} < \frac{1}{100}$
(or equivalent).

A response that states either $\text{Error} = \frac{0.3}{6}\frac{1}{2}^3$
or $\text{Error} = \frac{1}{160}$
does not earn the second point.

The student response accurately includes **both** of the criteria below.

Form of the error bound

Analysis

Solution:

$$P_2 1.5 - f 1.5 \leq \frac{\max_{1 \leq x \leq 1.5} f''' x}{3!} \quad 1.5 - 1^3 \quad \leq \frac{0.3}{6} \quad \frac{1}{2}^3$$

$$= \frac{1}{160} \quad < \frac{1}{100}$$